AI 504 final exam

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1 Instructions (read carefully before starting!)

This exam packet consists of:

1. Instructions for the exam
2. Multiple choice and free-response exam questions
3. Notes for your reference during the exam

There are 12 multiple choice and 8 free response questions on this exam. Develop your solutions to this exam in the text editor of your choice. **It must be very clear which solutions correspond to which problems, so carefully distinguish multiple choice vs. free response and carefully label each solution in each section with the number of the corresponding problem.** Additionally,

- There is *exactly one correct response* for each multiple choice question. (You're welcome!)
- *You can write your solutions in either English or Danish.* Please approximate mathematical symbols as best you can. That said, I do understand that formatting is not your priority during the exam.
- Besides the provided reference, no other aids are allowed.

Grading

Each section will be graded separately. Your final exam grade will be the average of your scores, rounded in the direction of the free response score. So for example if you get a 12 on the multiple choice and a 7 on the free response, your total exam grade is 7. If you get a 12 on the free response and a 7 on the multiple choice, your total exam grade is 10.

- Your multiple choice score is the number of questions you get right, rounded *up* to the next grade score. So 11 or 12 questions right gets a 12; 8, 9, or 10 questions right gets a 10, and so on.
- For the free response:
  - A score of 12 means *correct solutions* to at least 6 of the 8 questions. A correct solution allows for typos or other minor errors, but should contain all the necessary ideas, thoroughly and clearly explained with all sufficient justification, to solve the problem.
  - A score of 10 means correct solutions to at least 4 of the 8 questions, to the same standard as above.
  - A score of 7 means correct solutions to at least 2 of the 8 questions, to the same standard as above, plus partial solutions to at least 2 more questions. A *partial solution* may contain major errors, but **demonstrates an understanding of what the problem is asking,** plus makes **nontrivial progress towards a solution.**
  - A score of 4 means a partial solution to at least 4 of the 8 questions, to the same standard as above.
  - A score of 2 means a partial solution to at least 2 of the 8 questions.
  - Anything less will be awarded a 0.

2 Multiple-choice questions

1. Which of the following first-order sentences correctly translates the English phrase "the product of any two negative numbers is positive?"

   (a) $(\forall x)(\forall y)((x < 0 \vee y < 0) \to 0 < x \cdot y)$

   (b) $(\forall x)(\forall y)((x < 0 \wedge y < 0) \to 0 < x \cdot y)$

   (c) $(\forall x)(\forall y)(x \cdot y < 0 \to (x < 0 \vee y < 0))$

   (d) $(\forall x)(\exists y)((x < 0 \wedge y < 0) \to 0 < x \cdot y)$

   Answer: (b). The phrase "any two negative numbers" implies *both* numbers are negative, ruling out (a). (c) says that "if the product of two numbers is negative, then one of them must be negative," which is true but not what the question asks. (d) futzes with the quantifier; it is again true but fails to translate *any two*.

2. Which of the following first-order sentences correctly translates the $\mathcal{A}(\mathcal{RC})$-sentence All (fear all mammals) squirrels? (Rendered more naturally into English: "all who fear all mammals are squirrels.")

   In the first-order signature $\mathsf{fear}(a, b)$ means "a fears b," $\mathsf{squirrel}(a)$ means "a is a squirrel," and $\mathsf{mammal}(a)$ means "a is a mammal."

   (a) $(\forall y)((\forall x)(\mathsf{mammal}(x) \to \mathsf{fear}(y, x)) \to \mathsf{squirrel}(y))$

   (b) $(\forall x)(\forall y)(\mathsf{mammal}(x) \to (\mathsf{fear}(y, x) \to \mathsf{squirrel}(y)))$

   (c) $(\exists x)(\forall y)((\mathsf{mammal}(x) \wedge \mathsf{fear}(y, x)) \to \mathsf{squirrel}(y))$

   (d) $(\forall y)((\forall x)(\mathsf{mammal}(x) \to \mathsf{fear}(x, y)) \to \mathsf{squirrel}(y))$

   Answer: (a). The phrase $(\forall x)(\mathsf{mammal}(x) \to \mathsf{fear}(y, x))$ correctly translates "y fears all mammals," which is the key observation here. (b) incorrectly parenthesizes the implications and (c) and (d) mess up the quantifiers. (d) *looks like* it is just switching adjacent universal quantifiers, which would be equivalent, but this is not so: $(\forall y)$ and $(\forall x)$ are not actually adjacent because the latter is contained in the hypothesis of the implication.

3. A relation R is transitive in case $(\forall x)(\forall y)(\forall z)\big(((x, y) \in R \wedge (y, z) \in R) \to (x, z) \in R\big)$. Which one of the following first-order sentences is *not* also a correct definition of transitivity?

   (a) $(\forall y)(\forall z)(\forall x)\big(((x, y) \in R \wedge (y, z) \in R) \to (x, z) \in R\big)$

   (b) $(\forall x)(\forall z)\big((\exists y)((x, y) \in R \wedge (y, z) \in R) \to (x, z) \in R\big)$

   (c) $(\forall x)(\forall y)(\forall z)\big((x, y) \in R \to ((y, z) \in R \to (x, z) \in R)\big)$

   (d) $(\forall x)(\forall y)(\exists z)\big(((x, y) \in R \wedge (y, z) \in R) \to (x, z) \in R\big)$

   Answer: (d). (a) is perhaps the "standard definition," (c) uses the fact that $(p \wedge q) \to r$ is logically equivalent to $p \to (q \to r)$, and (b) uses the fact that we can push the $(\forall y)$ quantifier into the hypothesis of the implication, since it doesn't appear in the conclusions, and there it switches to $(\exists y)$. (d) simply changes $(\forall z)$ to $(\exists z)$ which certainly does not preserve logical equivalence.

4. Which of the following functions $f : \mathbb{R} \to \mathbb{R}$ satisfies the first-order sentence $(\exists y)(\forall x)(f(x) < y)$? (Here the quantifiers range over real numbers.)

   (a) $f(x) = -x$

   (b) $f(x) = 1$ (the constant-1 function)

   (c) $f(x) = x$

   (d) $f(x) = x^2$

Answer: (b). This condition simply says that there is a single y-value (think horizontal line) that the graph of the function f never crosses. Clearly functions (a), (c), and (d) all get arbitrarily large and their graphs do not lie entirely below any horizontal line.

5. Work over the propositional-logic signature $\{p, q, r\}$. How many models of $(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)$ are there?

   (a) 1
   (b) 2
   (c) 3
   (d) 4

   Answer: (b). This formula simply says that p, q, and r all have the same boolean value. So it is satisfied by the one that makes them all true, and the one that makes them all false.

6. Suppose φ and ψ are satisfiable propositional sentences. Which of the following sentences also must be satisfiable?

   (a) $\varphi \wedge \psi$
   (b) $\varphi \vee \psi$
   (c) $\varphi \leftrightarrow \psi$
   (d) $\varphi \rightarrow \neg\psi$

   Answer: (b). Any model satisfying either φ or ψ will satisfy $\varphi \vee \psi$, so even if only one formula were satisfiable then (b) would have to be satisfiable as well. On the other hand, none of the other answers work. For (a) and (c) take φ to be p and ψ to be $\neg p$; each is satisfiable but the combination isn't. For (d) take φ and ψ to be T, then (d) is equivalent to F which isn't satisfiable.

7. Which of the following statements is *not* equivalent to "φ is a tautology?" (Work in propositional logic.)

   (a) $\emptyset \models \varphi$
   (b) $\{\varphi\} \models \mathsf{T}$
   (c) $\{\neg\varphi\} \models \mathsf{F}$
   (d) $\{\mathsf{T}\} \models \varphi$

   Answer: (b), which says absolutely nothing about φ. (*Everything* entails T.) (a) says that every model satisfying the empty set of formulas satisfies φ, but *every* model satisfies all formulas in $\emptyset$ (vacuous reasoning), so it's the same as saying all models satisfy φ. (d) says that every model satisfying T (which is, again, all of them) satisfies φ. (c) says that every model satisfying $\neg\varphi$ must satisfy F. This means that *no* models satisfy $\neg\varphi$, which says that every model satisfies φ.

8. Which of the following sentences of propositional logic over the signature $\{p\}$ is a tautology?

   (a) p
   (b) $(p \rightarrow p) \rightarrow p$
   (c) $((p \rightarrow p) \rightarrow p) \rightarrow p$
   (d) $(((p \rightarrow p) \rightarrow p) \rightarrow p) \rightarrow p$

   Answer: (c). Obviously every sentence of this form is satisfied when p is T, so it remains to check which remain satisfied when p is F. Notice that $\varphi \rightarrow \mathsf{F}$ is equivalent to $\neg\varphi$, so each time we add an extra $\cdots \rightarrow \mathsf{F}$ it has the effect of flipping the truth value. So the question is: when we start with F, how can we end up at "true?" By flipping the truth value an odd number of times, which means (c).

9. Work over the signature $\{1, 2, 3, \dots\}$ in the logic $\mathcal{A}$. Let Γ be the set of sentences $\{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 3, \mathsf{All}\ 3\ 4, \dots\}$. Which of the following sentences does Γ entail?

   (a) All 3 1

   (b) All 300 1000

   (c) All 300 100

   (d) All 23 22

   Answer: (b). $\Gamma \models \mathsf{All}\ m\ n$ exactly when $m \leq n$.

10. Work over the $\mathcal{A}(\mathcal{RC})$-signature with a single noun p and a single verb r. Consider models where the domain is $\mathbb{R}$ (the real line) and $(x, y) \in [\![r]\!]$ iff $|x - y| < 1$, i.e., the distance between x and y is less than 1.

    For which interpretations of $[\![p]\!]$ does $[\![r\ \mathsf{all}\ p]\!]$ consist of a *single point*? Note that (a, b) is the open interval with endpoints a and b, i.e., $\{x \in \mathbb{R} : a < x < b\}$.

    (a) $[\![p]\!] = (0, \frac{1}{4})$

    (b) $[\![p]\!] = (0, \frac{1}{2})$

    (c) $[\![p]\!] = (0, 1)$

    (d) $[\![p]\!] = (0, 2)$

    Answer: (d). $[\![r\ \mathsf{all}\ p]\!]$ is simply the set of points which are distance at-most-one away from *all* points in $[\![p]\!]$. But for choices (a), (b), (c), it's easy to see that $[\![p]\!] \subseteq [\![r\ \mathsf{all}\ p]\!]$, as every point in $[\![p]\!]$ is distance at most one away from every other point in $[\![p]\!]$. (Said another way: if I pick any two points in $[\![p]\!]$, the distance between them is at most 1 for choices (a)-(c).) However for (d) it's only the point 1 which is distance at most 1 away from every other point in the interval (0,2).

11. Work over the $\mathcal{A}$-signature $\{1, 2\}$. (There are only four sentences in this signature!) Suppose φ and ψ are two sentences such that $\{\varphi\} \models \psi$ but $\{\psi\} \not\models \varphi$. What can we conclude?

    (a) $\varphi \in \{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 1\}$ and $\psi \in \{\mathsf{All}\ 1\ 1, \mathsf{All}\ 2\ 2\}$

    (b) $\psi \in \{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 1\}$ and $\varphi \in \{\mathsf{All}\ 1\ 1, \mathsf{All}\ 2\ 2\}$

    (c) Both $\varphi, \psi \in \{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 1\}$

    (d) Both $\varphi, \psi \in \{\mathsf{All}\ 1\ 1, \mathsf{All}\ 2\ 2\}$

    Answer: (a). If $\{\psi\} \not\models \varphi$ then $\emptyset \not\models \varphi$, which means that φ cannot be in the set $\{\mathsf{All}\ 1\ 1, \mathsf{All}\ 2\ 2\}$. This eliminates (d) and (b). On the other hand if both $\varphi, \psi \in \{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 1\}$, then $\{\varphi\} \not\models \psi$, which eliminates (c).

12. Work in a general logic. Suppose there is a sentence φ and two models $\mathcal{M}$ and $\mathcal{N}$ which disagree on φ (i.e., one satisfies it and one falsifies it), and suppose there is a sentence ψ such that $\{\psi\} \models \varphi$. Which of the following situations is *impossible?*

    (a) $\mathcal{M}$ falsifies ψ

    (b) $\mathcal{N}$ satisfies ψ

    (c) Both $\mathcal{M}$ and $\mathcal{N}$ satisfy ψ

    (d) $\mathcal{M}$ and $\mathcal{N}$ disagree on ψ

    Answer: (c). If both models satisfy ψ and $\{\psi\} \models \varphi$, then both models would have to satisfy φ as well. On the other hand both models could *falsify* ψ, which eliminates (a) and (d), or $\mathcal{N}$ could satisfy ψ and φ and $\mathcal{M}$ could falsify ψ and φ, which eliminates (b).

3 Free response questions

How's it going? I hope you're enjoying yourself!

1. Consider the following list of five sentences.

   (a) Sentences (e) and (b) disagree.

   (b) Sentences (a) and (c) disagree.

   (c) Sentences (b) and (d) disagree.

   (d) Sentences (c) and (e) disagree.

   (e) Sentences (d) and (a) disagree.

   A *solution* is an assignment of a boolean value to each of the five sentences such that conditions (a)-(e) are satisfied. That is to say, if the boolean value assigned to a particular sentence is *true*, the two adjacent sentences should *disagree,* i.e., have different truth values. If the boolean value assigned to a particular sentences is *false*, then the two adjacent sentences should *agree,* i.e., have the same truth value. **How many solutions are there, and what are they?**

   Answer. There are five sentence letters here $p_1, \ldots, p_5$. Sentence (i) says $p_i \leftrightarrow \neg(p_{i-1} \leftrightarrow p_{i+1})$ for $1 \leq i \leq 5$, and the question asks us to enumerate the models satisfying the conjunction of these 5 sentences. There are 32 models, which is probably small enough to brute-force it.

   However there is a more elegant solution. The trick is to express p_{i+1} as a function of p_{i-1} and p_i, instead of p_i as a function of p_{i+1} and p_{i-1}. Now if p_i is false, it says that p_{i+1} is the same as p_{i-1}, and if p_i is true, it says that p_{i+1} is different from p_{i-1}, which gives us the following table:

p_{i-1}	p_i	p_{i+1}
F	F	F
F	T	T
T	F	T
T	T	F

   The advantage of this is that we can p_1 and p_2 and then go around the circle applying this rule to the previous two values to get the next one. If we end back where we started, we got a satisfying model. So if p_1, p_2 is F,F, then we get the pattern F,F,F... and one satisfying model. If p_1, p_2 is any of the other initial values, we get a pattern repeating two T's and one F in a cycle of three, e.g, F,T,T,F,T,T,F,T,T,... — the only difference is where in the cycle we start. But this will not end up in a satisfying model unless the number of sentence letters is a multiple of three. So in the end, we get **one satisfying model—all false.**

2. Work in the logic $\mathcal{A}(\mathcal{RC})$ and over the signature with two nouns, `c` and `d`, and one verb `r`.

   Let $\mathcal{M}$ be any model in this signature satisfying the following two properties:

   - The denotation of any term that ends in `d` is nonempty, i.e., $[\![\texttt{d}]\!]$, $[\![\texttt{r all d}]\!]$, $[\![\texttt{r all (r all d)}]\!]$, ... are all nonempty subsets of M.
   - $[\![\texttt{r}]\!]$ is a transitive relation, i.e., for any $x, y, z \in M$, if $(x, y) \in [\![\texttt{r}]\!]$ and $(y, z) \in [\![\texttt{r}]\!]$ then $(x, z) \in [\![\texttt{r}]\!]$.

   (a) Suppose that $\mathcal{M} \models$ `All c (r all (r all d))`. Prove that $\mathcal{M} \models$ `All c (r all d)`.

   Hint. $\mathcal{M} \models$ `All c (r all d)` means $(\forall x \in [\![\texttt{c}]\!])(\forall y \in [\![\texttt{d}]\!])\,(x, y) \in [\![\texttt{r}]\!]$. So fix $x \in [\![\texttt{c}]\!]$ and $y \in [\![\texttt{d}]\!]$; now your goal is to prove $(x, y) \in [\![\texttt{r}]\!]$.

   (b) Construct a model $\mathcal{N}$ in the same signature that does *not* satisfy these properties, such that $\mathcal{N} \models$ `All c (r all (r all d))` but $\mathcal{N} \not\models$ `All c (r all d)`.

   (*Hint.* There are tiny models which work. List the elements of the domain and the denotations of `c`,`d`,`r`. From these derive the denotations of `r all d` and `r all (r all d)`.)

Answer. For part (a), fix $x \in [\![c]\!]$ and $y \in [\![d]\!]$ as indicated. By the first bullet point, there is $z \in [\![\mathtt{r}\ \mathsf{all}\ \mathtt{d}]\!]$. And since $\mathcal{M} \models \mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d}))$, $x \in [\![\mathtt{r}\ \mathsf{all}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})]\!]$. In particular this means that $(x, z) \in [\![r]\!]$. Since $z \in [\![\mathtt{r}\ \mathsf{all}\ \mathtt{d}]\!]$ and $y \in [\![d]\!]$, $(z, y) \in [\![r]\!]$. Since (x, z) and $(z, y) \in [\![r]\!]$, $(x, y) \in [\![r]\!]$, as the relation is transitive. But this is what we wanted to prove.

For part (b) simply let $\mathcal{N}$ be any model in which $[\![\mathtt{c}]\!]$, $[\![\mathtt{d}]\!]$ are nonempty subsets of the domain and $[\![r]\!]$ is the empty relation. Then $[\![\mathtt{r}\ \mathsf{all}\ \mathtt{d}]\!]$ is empty whereas $[\![\mathtt{r}\ \mathsf{all}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})]\!]$ is the entire domain. This means that $\mathcal{N} \models \mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d}))$ but $\mathcal{N} \not\models \mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})$.

3. Work in the logic $\mathcal{A}(\mathcal{RC})$ and over the signature with two nouns, c and d, and one verb r.

   It is *not* true that $\{\mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})\} \vdash \mathsf{All}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{c})\ \mathtt{d}$. For example, we could observe that $\{\mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})\} \not\models \mathsf{All}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{c})\ \mathtt{d}$ (there are very simple counter-models), then use soundness.

   The point of this exercise is to establish $\{\mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})\} \nvdash \mathsf{All}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{c})\ \mathtt{d}$ using *entirely proof-theoretic methods,* and avoid discussion of models entirely. **Your task is therefore to define a property of formulas** such that:

   - $\mathsf{All}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{c})\ \mathtt{d}$ *has* this property,
   - $\mathsf{All}\ \mathtt{c}\ (\mathtt{r}\ \mathsf{all}\ \mathtt{d})$ does *not have* this property,
   - *no* sentence of the form "$\mathsf{All}\ x\ x$" has this property, and
   - if $\Gamma \vdash \varphi$ and φ has this property, then at least one sentence in Γ has this property. (This will use induction on proof trees in $\mathcal{A}(\mathcal{RC})$, it is your choice whether to work with the rule DOWN or ANTI.)

   Define such a property of formulas and prove each of these four bullet points. There are *really simple* properties that will work!

   Answer. For example, we could define a property of $\mathsf{All}\ s\ t$ like "r occurs in s but not t." This clearly satisfies the first three bullet points. For the last, we need to show that if a sentence with this property is the conclusion of one of the proof rules, then at least one of the premises must have this property too. I'm going just write the crucial observation here.

   Here's the neat thing: a sentence with this property *cannot* follow from either ANTI or DOWN, because *any conclusion of those rules must have a verb in the second term!* So the only possible rule such a sentence can follow from is BARBARA!

   OK, so suppose $\mathsf{All}\ s\ t$ is the conclusion of a BARBARA rule and has the property that r occurs in s but not t. Then there is a term x such that the hypotheses are $\mathsf{All}\ s\ x$ and $\mathsf{All}\ x\ t$. Break up into cases depending on whether x contains an occurrence of r or not. If so, then $\mathsf{All}\ x\ t$ satisfies this property (an occurrence of r in the first term but not the second). If not, then $\mathsf{All}\ s\ x$ satsfies this property. In either case *one* of the premises satisfies this property, so we're done.

4. Work in the logic $\mathcal{A}$ and the signature 1,2,3. Suppose φ is a formula such that

   - $\{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 3\} \models \varphi$ and
   - $\{\mathsf{All}\ 3\ 2, \mathsf{All}\ 2\ 1\} \models \varphi$.

   What are the possible choices for φ? Answer, making sure to thoroughly justify why the sentences you gave are the only ones which work.

   Answer. Suppose φ is $\mathsf{All}\ x\ y$ for $x, y \in \{1,2,3\}$. If $x < y$ then $\{\mathsf{All}\ 3\ 2, \mathsf{All}\ 2\ 1\} \not\models \varphi$. If $x > y$ then $\{\mathsf{All}\ 1\ 2, \mathsf{All}\ 2\ 3\} \not\models \varphi$. Hence $x = y$, in which case both bullet points are clearly satisfied. So we only have the 3 sentences of the form $\mathsf{All}\ x\ x$ for $x \in \{1,2,3\}$.

5. Work in the logic $\mathcal{A}$, over any signature. We say a model $\mathcal{M}$ in the signature P is *fluffy* in case $[\![p]\!]$ is nonempty for every noun $p \in P$. Define $\Gamma \models^f \varphi$ in case every fluffy model satisfying Γ must also satisfy φ.

   Now it's certainly true that if $\Gamma \models \varphi$ then $\Gamma \models^f \varphi$ as well. (This has nothing to do with fluffiness, it's just a general fact that entailment over a restricted class of models is a more permissive relation.) But for this question you must **prove the converse,** that is, assuming that $\Gamma \models^f \varphi$, prove that $\Gamma \models \varphi$.

   (Note that unlike examples we saw in class and the practice problems where the converse relation was *false*, here the converse relation is *true.* Your hint is, given an arbitrary model, can you make it fluffy without changing the sentences it satisfies or falsifies?)

   Answer. Assume that $\Gamma \models^f \varphi$. Our goal is to show $\Gamma \models \varphi$. So suppose $\mathcal{M}$ is an arbitrary model that satisfies Γ. Our goal is now to show that $\mathcal{M}$ satisfies φ.

   $\mathcal{M}$ may not be fluffy. But we can easily define a fluffy "one-point extension" of $\mathcal{M}$ as follows. In English: we add a single point to the domain of $\mathcal{M}$, and we add this new point to the denotations of each of the nouns as well. In mathematics: our new domain is $M \cup \{\star\}$, where $\star$ is a new point. The denotation of p in the new model is the denotation of p in the old model plus the point $\star$, for each noun p. Call this model $\mathcal{M}^\star$. It is fluffy because the denotation of every sentence letter contains at least this new point $\{\star\}$.

   Notice that $\mathcal{M}$ and $\mathcal{M}^\star$ satisfy *exactly the same sentences*! This is because for any two sets X and Y and any point $\star$ not in either of them, $X \subseteq Y \iff X \cup \{\star\} \subseteq Y \cup \{\star\}$. Hence $\mathcal{M}^\star \models \Gamma$. But since $\Gamma \models^f \varphi$, $\mathcal{M}^\star \models \varphi$. But then $\mathcal{M} \models \varphi$, which is what we wanted to show in the first place.

6. Work in a general logical system with a notion of model and proof that has both soundness and completeness. Let $\mathcal{M}_\Gamma$ be the canonical model of a set of sentences Γ. This means that $\Gamma \vdash \varphi \iff \mathcal{M}_\Gamma \models \varphi$. Is it also true that $\Gamma \models \varphi \iff \mathcal{M}_\Gamma \models \varphi$? There are two parts to this question:

   (a) If $\Gamma \models \varphi$, must $\mathcal{M}_\Gamma \models \varphi$? Either prove or give a counterexample.

   (b) If $\mathcal{M}_\Gamma \models \varphi$, must $\Gamma \models \varphi$? Either prove or give a counterexample.

   Again, we can work entirely on the level of a general logic as outlined in the first page of this exam. Remember you can assume this logic has soundness and completeness.

   Answer. To part (a)—yes. This has nothing to do with canonicity specifically, and applies to all models of Γ. Since $\mathcal{M}_\Gamma \models \Gamma$ and every model of Γ also models φ, $\mathcal{M}_\Gamma \models \varphi$.

   To part (b)—yes as well, but now we use canonicity. Suppose $\mathcal{M}_\Gamma \models \varphi$. Then, by property of being a canonical model, $\Gamma \vdash \varphi$, and by soundness, $\Gamma \models \varphi$.

7. We define the *purely implicational* sentences of propositional logic as follows:

   - Any single sentence letter is purely implicational. (For the purposes of this question, we forbid the boolean constants T,F.)
   - If φ and ψ are purely implicational sentences, so is $\varphi \to \psi$.

   Intuitively speaking, it's just those sentences of propositional logic which do not have boolean constants and do not use any other boolean operator.

   Now we define the *reversal* function on purely implicational sentences by recursion as follows:

   - $p^R = p$, where p is a single sentence letter, and
   - $(\varphi \to \psi)^R = \psi^R \to \varphi^R$.

   So, for example, $((p \to r) \to (p \to q))^R = (q \to p) \to (r \to p)$.

   My question is, for any purely implicational sentence φ, **if φ is a tautology, must φ^R also be a tautology?** Either prove (for any such φ, if φ is a tautology then so is φ^R) or give a counterexample (a particular sentence φ such that φ is a tautology but φ^R is not).

   Answer. No. For example, let φ be $p \to (q \to p)$. This is a tautology (check), but φ^R is $(p \to q) \to p$, which is falsified by taking p, q to be F,T.

8. Express the relation $\min(a, b) < \max(c, d)$ by a first-order formula in the variables a, b, c, d that only uses the relation $<$. Explain carefully why your answer is correct.

   (*Hint.* For example, $x < \min(y, z) \iff (x < y) \wedge (x < z)$. In other words, a number is less than the minimum of two numbers iff it's less than *both of them.*)

   Answer. $\min(a, b) < \max(c, d)$ is satisfied exactly when *one of* $\{a, b\}$ is less than *one of* $\{c, d\}$. In other words,

   $$a < c \vee a < d \vee b < c \vee b < d.$$

Reference notes

4 General notions

A *logic* generally have the following working parts: signature, sentence, model, proof. The latter three are *indexed* by signatures so that, e.g., for each signature Σ, we have Σ-sentences, Σ-models, and Σ-proofs. Henceforth when we combine sentences, models, and proofs, we generally assume that we are working within a common signature. (If not we would get "type errors.")

4.1 Entailment and derivability

What models, proofs, and sentences are vary from logic to logic. However, they all have the following common features.

- Any model either satisfies or falsifies any given sentence. We write $\mathcal{M} \models \varphi$ and $\mathcal{M} \not\models \varphi$ respectively.
- Any proof has an associated set of *axioms* and a *conclusion*; these are all sentences.

From these basic notions we get:

- *Entailment:* $\Gamma \models \varphi$ means that every model satisfying (all sentences in) Γ must also satisfy φ.
- *Derivability:* $\Gamma \vdash \varphi$ means that there exists a proof whose associated set of axioms is a subset of Γ and whose conclusion is φ. (Note that the set of axioms need only be a subset of, not equal to, Γ!)

Warning: the double turnstile symbol $\models$ is used in two different senses: this model satisfies this sentence vs. this set of sentences entails this sentence. This is somewhat unfortunate, but more or less baked into practice. Disambiguate using context, and do not get confused!

Note: the definition of $\Gamma \models \varphi$ is **universal** (every model ...) whereas the definition of $\Gamma \vdash \varphi$ is **existential** (there exists a proof ...).

4.2 Soundness and completeness

A notion of derivability $\vdash$ is *sound* relative to a notion of entailment $\models$ if for any set of sentences Γ and single sentence φ, $\Gamma \vdash \varphi \implies \Gamma \models \varphi$. Conversely it is *complete* in case $\Gamma \models \varphi \implies \Gamma \vdash \varphi$. Both directions tend to be nontrivial as they relate existential and universal properties.

If we unpack the "soundness" direction, it says "if there exists a proof of φ from Γ, then every model satisfying Γ must satisfy φ. Pulling out the existential quantifier out of the hypothesis of the implication, we get a universal quantifier. So it says something like, "for all proofs T, if T is a proof of φ from Γ, then every model satisfying Γ must satisfy φ." Since this is a universal statement about proofs, and proofs tend to have some recursive structure (like a tree), we often use induction to prove soundness.

The single technique we have studied to prove completeness is that of a *canonical model*. Given a set of sentences Γ, a canonical model $\mathcal{M}_\Gamma$ is a model which satisfies "only those sentences that it has to," i.e., only those sentences that are derivable from Γ. So it has the characteristic property that $\mathcal{M}_\Gamma \models \varphi \iff \Gamma \vdash \varphi$. **You should be able to verify** that existence of a canonical model implies completeness.

Of course we cannot just stipulate such a model by fiat, we actually have to define it based on what our logic says a model is. The construction of a canonical model is usually one of the trickier definitions associated with each logic, and the proof that is has this characteristic property one of the trickier proofs.

5 Specific logics

5.1 $\mathcal{A}$

- *Signatures* are simply sets; we call them "nouns," but this is just how we think of them—it has no mathematical substance.

- Given a signature P, a *P-model* is a set M together with an interpretation $[\![p]\!] \subseteq M$ for each $p \in P$.
- Similarly a *P-sentence* is a pair $\mathsf{All}\ p\ q$. Mathematically this is no different from the ordered pair (p, q).
- $\mathcal{M} \models \mathsf{All}\ p\ q \iff [\![p]\!] \subseteq [\![q]\!]$.
- A proof in $\mathcal{A}$ of φ from Γ is a binary tree whose root is labeled by φ and each of whose leaves is either in Γ, or has the form $\mathsf{All}\ x\ x$ for some $x \in P$. Furthermore each non-leaf must have the form

$$\frac{\mathsf{All}\ x\ y \qquad \mathsf{All}\ y\ z}{\mathsf{All}\ x\ z}$$

  for some $x, y, z \in P$. This rule is called "barbara."
- If you're proving something by induction your base cases are (i) what if your tree is a single leaf of the form $\mathsf{All}\ x\ x$ and (ii) what if it's a single leaf labeled by an element of Γ? Otherwise, your proof is an instance of barbara applied to two sub-proof trees.
- Relative to the notions of model and proof above, this logic is complete. The canonical model $\mathcal{M}_\Gamma$ has domain P and $[\![p]\!] = \{x \in P : \Gamma \vdash \mathsf{All}\ x\ p\}$. Notice P plays the role of both signature and domain of $\mathcal{M}_\Gamma$; don't let this confuse you.

5.2 $\mathcal{A}(\mathcal{RC})$

- *Signatures* are an (ordered) pair of sets, one of which we call "nouns," the other of which we call "verbs." We typically use P and R respectively for these.
- Given a signature P, a *P-model* is a set M together with an interpretation $[\![p]\!] \subseteq M$ for each $p \in P$ and $[\![r]\!] \subseteq M \times M$ for each $r \in R$.
- A (P, R)-*term* is either a single noun (base case) or "$r\ \mathsf{all}\ t$," where t is a term and $r \in R$. Think of this as a nonempty list where the last element is a noun and all the rest are verbs. Given a model $\mathcal{M}$, each term t has an interpretation $[\![t]\!] \subseteq M$; this is given by the interpretation of the nouns in the base case and $[\![r\ \mathsf{all}\ t]\!] = \{x \in M : \forall y \in [\![t]\!]\ (x, y) \in [\![r]\!]\}$.

  Term induction: to show that all (P, R)-terms have some property, first show that individual nouns have that property, then show that if any term t has that property, so does $(r\ \mathsf{all}\ t)$, for any verb r.
- A (P, R)-*sentence* is a pair $\mathsf{All}\ s\ t$, where s, t are (P, R)-terms.
- $\mathcal{M} \models \mathsf{All}\ s\ t \iff [\![s]\!] \subseteq [\![t]\!]$.
- A proof in $\mathcal{A}(\mathcal{RC})$ is a tree labeled by sentences. Just like in $\mathcal{A}$, base cases are either sentences in Γ or sentences which have the form $\mathsf{All}\ x\ x$, for some term x. We still have the rule "barbara." But we actually define *two* new proofs systems, one by adding the rule "down,"

$$\frac{\mathsf{All}\ x\ (r\ \mathsf{all}\ \mathrm{y}) \qquad \mathsf{All}\ z\ y}{\mathsf{All}\ x\ (r\ \mathsf{all}\ z)}$$

  for some terms x, y, z and verb r; and the other by adding the rule "anti,"

$$\frac{\mathsf{All}\ x\ y}{\mathsf{All}\ (r\ \mathsf{all}\ y)\ (r\ \mathsf{all}\ x)}$$

  for some terms x, y and verb r. Even though these are two different notions of proof, they induce the **same notion $\vdash$ of derivability,** and you should understand what this means and be able to explain why.

- Relative to the notions of model and proof above, this logic is complete. The domain of canonical model $\mathcal{M}_\Gamma$ is the set of all (P,R)-terms; $[\![p]\!] = \{x \in (P,R)\text{-terms} : \Gamma \vdash \mathsf{All}\ x\ p\}$ and $[\![r]\!] = \{(x,y) \in (P,R)\text{-terms} \times (P,R)\text{-terms} : \Gamma \vdash \mathsf{All}\ x\ (r\ \mathsf{all}\ y)\}$.

  One then has to prove that for all (P,R)-terms t, $[\![t]\!] = \{x \in (P,R)\text{-terms} : \Gamma \vdash \mathsf{All}\ x\ t\}$. From this you can prove the canonical model property relatively easily.

5.3 Propositional logic

- Signatures are just single sets, which we call the set of "sentence letters."
- Given a signature P, a P-model is a function $[\![\cdot]\!] : P \to \mathsf{Bool}$.
- A P-formula is either an element of p or a boolean constant T or F (base cases), $\varphi \star \psi$ where φ, ψ are P-formulae and $\star \in \{\wedge, \vee, \to, \leftrightarrow\}$, or $\neg\varphi$ where φ is a P-formula.
- Given a model $\mathcal{M}$ and a formula φ, we define $[\![\varphi]\!] \in \mathsf{Bool}$ by induction on the construction of φ. If φ is a single letter $p \in P$, then $[\![\varphi]\!]$ is $[\![p]\!]$, which is part of the data for $\mathcal{M}$. If φ is T/F, then $[\![\varphi]\!]$ is T/F respectively. If φ is $\theta \star \psi$ for some $\star \in \{\wedge, \vee, \to, \leftrightarrow\}$, or φ is $\neg\theta$, then $[\![\varphi]\!]$ is obtained from $[\![\theta]\!]$ and $[\![\psi]\!]$ by applying the particular rules for $\star$.

  (So $[\![\theta \vee \psi]\!] = \mathsf{T}$ just in case either $[\![\theta]\!] = \mathsf{T}$ or $[\![\psi]\!] = \mathsf{T}$, and so on. You should be familiar with the meanings of each of the connectives.)
- A formula is a **tautology** in case it is satisfied by every model, **satisfiable** in case it is satisfied by some model, **falsifiable** in case it is falsified by some model, and a **contradiction** in case it is falsified by every model.
- A *tableau* is a tree each of whose vertices is labeled by two sets of formula: the true ones and the false ones. For each non-leaf, its children are determined by picking some formula in one of the two sets and simplifying it. So for example if $T : \varphi \vee \psi$, we split into two children, one with $T : \varphi$ and the other with $T : \psi$. If $F : \varphi \vee \psi$, then there is one child with $F : \varphi, \psi$. The other formulas are inherited from the parent. (You should be familiar with all the rules; there is one for each connective and each choice of $\{T, F\}$.)

  Each leaf is either inconsistent (there is overlap between the T- and F-sets), or contains no connective left to break down.
- To check whether $\Gamma \vdash \varphi$, make a tableau starting from all the sentences in Γ as the T-set, and the F-set just consisting of φ. Then complete the tableau. If any leaf is consistent, this gives us a counter-model, which satisfies every formula of Γ but falsifies φ. However if all leaves are inconsistent, this verifies $\Gamma \vdash \varphi$.

5.4 First-order logic

- Signatures in first-order logic are collections of constant, function, and relation symbols. Each function and relation symbol comes with an arity (a number of arguments that it takes). Sometimes we also want to distinguish different types.
- A model for each signature consists of a domain for each type, plus an interpretation of function, relation, and constant symbols by functions, relations, and constants of the correct type.
- In first-order logic, the main skills are to be able to **formalize natural language and informally evaluate whether a model satisfies a formula.**